



Project-Presentation

Course: [ME444] – Analysis and Design of Mechanical Systems

(Department of Mechanical Engineering)

Design and Experimental study of a Tuned Vibration Absorber System

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INTRODUCTION

Problem Statement

- Mechanical systems often face excessive vibrations at resonance
- This leads to:
 - Reduced performance
 - Structural damage
 - Noise and instability

Objective

- To design and build a vibration absorber system
- To reduce vibration amplitude at resonance
- To experimentally verify vibration control

METHODOLOGY

System Description

- A 2-DOF spring-mass system is developed
- Components:
 - Main mass + spring (primary system)
 - Absorber mass + spring (secondary system)
 - Motor with eccentric mass (excitation)

Working Principle

- System is excited using rotating eccentric mass
- At resonance → large vibrations
- Absorber is tuned to same frequency
- Vibration transfers to absorber → main mass stabilizes

WORK PLAN

Approach

1. Design frame and select components
2. Fabricate structure and assemble system
3. Apply forced vibration using motor
4. Add absorber and tune system
5. Perform experiments and record data
6. Plot graphs and analyze results

PERMISSIONS AND RESOURCES

Permissions

- Workshop access for fabrication of the frame/base.

Resources Required

- Acrylic base and frame
- Basic mounting equipment for the frame
- Springs and masses
- DC motor and power supply
- Measuring tools (scale, IMU)
- Microcontroller

EXPECTED OUTCOMES AND TIMELINES

Expected Outcomes

- Identification of resonance condition
- Reduction in vibration amplitude using absorber
- Observation of:
 - Resonance peak
 - Peak splitting (with absorber)
 - Effect of damping

Timeline

- **Week 1:** Design + material procurement
- **Week 2:** Frame fabrication
- **Week 2:** Main system assembly
- **Week 3:** Absorber installation
- **Week 3:** Testing and tuning
- **Week 3:** Data collection
- **Week 4 & 5:** Analysis and report